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Artificial Intelligence in Skin Cancer Diagnosis: A Reality Check

JID Open

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The field of skin cancer detection offers a compelling use case for the application of artificial intelligence (AI) within the realm of image-based diagnostic medicine. Through the analysis of large datasets, AI algorithms have the capacity to classify clinical or dermoscopic images with remarkable accuracy. Although these AI-based applications can operate both autonomously and under human supervision, the best results are achieved through a collaborative approach that leverages the expertise of both AI and human experts. However, it is important to note that most studies focus on assessing the diagnostic accuracy of AI in artificial settings rather than in real-world scenarios. Consequently, the practical utility of AI-assisted diagnosis in a clinical environment is still largely unknown. Furthermore, there exists a knowledge gap concerning the optimal use cases and deployment settings for these AI systems as well as the practical challenges that may arise from widespread implementation. This review explores the advantages and limitations of AI in a variety of real-world contexts, with a specific focus on its value to consumers, general practitioners, and dermatologists.

Keywords: Convoluted neural network, Melanoma, Dermoscopy, Mobile apps, Primary care

Journal of Investigative Dermatology (2024) **144,** 492−499; doi:10.1016/j.jid.2023.10.004

INTRODUCTION

The first instance of a writer speculating that humans could be replaced by intelligent machines is often attributed to Aristotle. In a section of his work *Politics*, he states “For if each of the tools were able to complete its own work when commanded, or by perceiving in advance, (…) master-craftsmen would need no subordinates nor masters of slaves” (Lagrandeur, 2020). Since the time of Aristotle, humanity has experienced continual technological evolution, progressing from simple tools to modern computers. In the dawn of the computer age, researchers initially believed that they could achieve meaningful results through rule-based systems, much like the syllogisms proposed by Aristotle. In recent years, however, data-driven artificial intelligence (AI) has gained prominence over rule-based AI.

Modern machine learning techniques rely on vast datasets to identify patterns useful for classification. Diagnostic imaging represents one of the most promising arenas for AI research. Skin cancer detection in particular serves as an appealing application for AI, given that diagnoses often hinge on the subjective visual interpretation of clinical and dermoscopic images. AI-assisted diagnosis promises several advantages. For instance, AI could improve access to specialist-level expertise. The scarcity of dermatologists is a serious problem in many regions, often leading to protracted waiting times for specialist appointments. In addition, there is growing optimism that AI-based systems might offer greater consistency and higher accuracy than human experts. Esteva et al (2017) first demonstrated the efficacy of convolutional neural networks (CNNs) for the task of image-based classification in dermatology. CNNs are specialized types of neural networks that are optimally suited for image analysis and are predominantly trained using supervised learning techniques. This requires that the images of the training dataset must be labeled with a diagnosis, serving as the ground truth. This label is essential for the system to learn the relationship between the input data and the corresponding diagnosis. Esteva et al (2017) demonstrated that an AI trained in this manner could differentiate between malignant and benign skin lesions at an expert level. Subsequent studies, such as those by Haenssle et al (2020) and Tschandl et al (2019b), corroborated these findings across different image sets and categories of skin disease. Fueled by advances in computing power and the increasing availability of training images, the diagnostic accuracy of AI systems continued to improve over time. However, the practical utility of these AI systems in real-world settings remains a subject of ongoing inquiry because most studies have been conducted in artificial environments. In this review, we explore AI applications in various real-world scenarios, shedding light on both their strengths and limitations.

AI TOOLS FOR CONSUMERS

Over recent decades, numerous smartphone apps designed for self-evaluation of skin lesions have emerged. More than

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Abbreviations: 3D, 3-dimensional; AI, artificial intelligence; BCC, basal cell carcinoma; CI, confidence interval; CNN, convolutional neural network; DEJ, dermal−epidermal junction; GP, general practitioner; LC-OCT, line-field confocal optical coherence tomography; OCT, optical coherence tomography; RCM, reflectance confocal microscopy; SCC, squamous cell carcinoma; TBP, body photography

Received 8 June 2023; revised 8 September 2023; accepted 1 October 2023; corrected proof published online 18 November 2023

half of these apps equip users with self-monitoring features, including the ability to log, organize, and track moles using the built-in camera on their mobile devices. Some apps serve as intermediaries, transmitting images to qualified experts for evaluation of potential risks on the basis of the submitted photos. Other apps autonomously categorize lesions as either high or low risk, offering guidance on whether medical consultation may be necessary.

In 2018, the Cochrane Skin Cancer Diagnostic Test Accuracy Group conducted a systematic review of studies published up to August 2016 examining the diagnostic accuracy of smartphone apps for identifying melanoma and borderline lesions (Chuchu et al, 2018). Their review highlighted the overall low methodological quality and scarcity of robust evidence in this area. Of the 1051 studies assessed, 203 were reviewed, 16 were identified as potentially eligible, and ultimately, only 2 were included. The primary reasons for exclusion were inappropriate study population, inappropriate index tests, absence of categorical outcome measures in the form of 2 × 2 tables, and the study being derivative in nature. Among the 2 studies that were included, one was a monocentric retrospective case-control study that aimed to analyze the accuracy of 4 undisclosed mobile apps in diagnosing 188 skin lesions (Wolf et al, 2013). The other was a prospective trial focused on testing the diagnostic capabilities of the SkinVision App in 144 lesions (Maier et al, 2015). The reference standard for both studies was the histopathological report. Both studies were considered to have high risks of bias and applicability concerns regarding population selection. They were conducted in specialized centers, which differed from the community setting where these apps are intended for use. In addition, many lesions were excluded owing to poor image quality, an issue that should be considered in real-world applications. Both studies also received high-risk ratings for bias concerning flow and timing. Across the 4 AI-based apps tested in these studies, true positive rates ranged from 7% (95% confidence interval [CI] = 2−16) to 73% (95% CI = 52−88), and true negative rates ranged from 37% (95% CI = 29−46) to 94% (95% CI = 87−97). The authors concluded that smartphone apps using AI-based analysis did not show sufficient accuracy and that they were associated with a high likelihood of missing melanomas (Chuchu et al, 2018).

Similar conclusions were reached in a more recent systematic review that also focused on the diagnostic accuracy of AI-powered smartphone apps for detecting skin cancer (Freeman et al, 2020). In addition to the 2 studies analyzed by Chuchu et al (2018), the authors included 5 more studies published after 2016 as well as 2 studies excluded by Chuchu et al (2018) owing to the low number of lesions included (Chadwick et al, 2014; Robson et al, 2012). Nearly all the included studies were prospective, and only 2 studies were retrospective (Chadwick et al, 2014; Wolf et al, 2013). One study broadened its focus beyond melanocytic lesions and included other types of skin cancers such as basal cell carcinoma (BCC) and squamous cell carcinoma (SCC) (Thissen et al, 2017). Three of the studies used expert assessment as the reference standard (Chung et al, 2018; Nabil et al, 2017; Ngoo et al, 2018), a criterion considered to be at high risk for bias. Only 2 trials were rated as being at low risk for patient selection bias (Dorairaj et al, 2017; Thissen et al, 2017), whereas all the studies were deemed to have a high risk of bias concerning flow and timing, patient selection, and index tests. One significant critique concerning all included studies was that the image quality was likely better than what would be expected in real-world settings. In retrospective studies, archived images were selected on the basis of their quality, whereas in prospective studies, clinicians acquired images using a standard protocol under optimized conditions with a single camera rather than participants using their own devices. It is worth noting that only 2 (TeleSkin skinScan and SkinVision) of the 6 mobile apps mentioned in this systematic review were still available at the time the manuscript was drafted. Two had been withdrawn from the market after investigations by the American Federal Trade Commission investigations, and 2 others were no longer available. Although no peer-reviewed studies evaluating the TeleSkin skinScan app could be located, the review authors did provide an analysis of the sensitivity and specificity of the SkinVision app, conducting a per-lesion assessment across various trials. As reported by Thissen et al (2017) in their study, the top-performing app exhibited a sensitivity of 88% and a specificity of 79%. In a hypothetical population of 1000 adults, assuming a melanoma prevalence of 3%, this level of performance would result in 4 of 30 melanomas going undetected. In addition, over 200 individuals would be burdened with false-positive results. The authors concluded that the poor performance was likely due to the low methodological quality of the included studies. Specific issues highlighted included selective participant recruitment, inadequate reference standards, differential verification processes, and a high frequency of images that could not be evaluated.

The SkinVision app was also assessed in a single-center, comparative, observational cohort study carried out in Switzerland from January to June 2021 (Jahn et al, 2022). In this study, patients underwent comprehensive examinations conducted by dermatologists, which included total body photography (TBP). In addition, all nevi measuring 3 mm or larger as well as any smaller melanocytic lesions deemed suspicious were evaluated using smartphones equipped with the specialized app. Of 1204 pigmented skin lesions analyzed, the smartphone app classified 980 (81%) lesions as benign and flagged 224 (19%) as carrying an increased risk for melanoma. In contrast, dermatologists diagnosed 1195 (99.3%) lesions as benign and identified only 9 (0.7%) as suspicious. As a result, the app classified pigmented skin lesions 27 times more frequently as suspicious than did the dermatologists, indicating a concerning rate of false positives.

Since 2020, the landscape of algorithm-based apps for skin cancer detection has been rapidly evolving, in part owing to significant improvements in the processing power and camera quality of mobile devices. Sun et al (2022) identified 25 apps focused on skin cancer self-detection by searching the Apple App Store, Google Play Store, and Google Search. They then tested these apps using a small independent set of clinical images comparable with those typically submitted through smartphones. The study highlighted considerable variability in the apps' outputs, including diagnosis, risk category, and risk score, with mean diagnostic accuracies of 56, 60, and 64%, respectively. The authors concluded that although a direct comparison was challenging, the accuracy of these apps was highly variable and generally low. Such inconsistencies could potentially lead to false reassurance and reluctance to seek medical care or, conversely, to emotional distress and increased healthcare utilization. However, a significant drawback of this study lies in its retrospective design. Specifically, if the mobile app under test did not permit image uploads, images were displayed on a 4K ultrahigh-definition monitor and then captured through smartphone, a method far removed from the real-life usage.

In a separate study, 2 new market-approved AI algorithms designed to detect and analyze skin tumors were evaluated in a single-center, prospective validation trial. Conducted in a tertiary referral center in Austria between June 2018 and December 2019 (Kränke et al, 2023), the study employed 5 different mobile phones. A total of 1171 lesions were included in the analysis. When classifying the lesions into benign versus nonbenign categories, the algorithm designed for analysis showed a sensitivity of 95.4% (95% CI = 93.5−97.3), whereas the algorithm focused on detection had a sensitivity of 96.4% (95% CI = 93.5− 98.9). Specificity was 90.3% (95% CI = 88.1−92.5) for the analysis and 94.9 (95% CI = 92.5−97.2) for the detection algorithm. Although these results are promising, the study authors noted that the potential for limited applicability may arise when the individuals selecting and capturing the images are patients rather than clinicians.

In conclusion, although there has been notable progress in the regulation of consumer apps designed for skin cancer diagnosis, there remains an ongoing requirement for stronger clinical validation to substantiate their purported benefits. These potential benefits are that consumer apps could offer immediate preliminary assessments right from the comfort of one's home, thus increasing access to healthcare services for those who may be limited by geography, time, or financial constraints. Furthermore, these apps could encourage regular self-skin examinations, promoting early detection and more proactive healthcare habits. However, utilizing consumer-generated images of skin lesions for skin cancer detection without proper control could give rise to various risks rooted in the possible absence of expertise, standard procedures, and supervision usually offered by clinicians.

AI AS A TOOL FOR GENERAL PRACTITIONERS

In theory, the incorporation of AI in primary care, when equipped with sufficient sensitivity, holds the potential to enhance the effective triage of high-risk lesions toward secondary care. This would ensure that individuals with skin cancer promptly receive appropriate medical attention. Conversely, a high specificity could minimize unnecessary referrals and promptly alleviate patient anxiety. Likewise, the fusion of AI with teledermatology could enable the transmission of referral urgency to specialized clinics or dermatologists supplemented by relevant clinical information and images. This integration has the potential to augment the efficiency and precision of these referral processes. However, it is crucial to maintain a high level of specificity to avoid unacceptable spikes in referral rates to dermatologists and specialized centers (Ferrante di Ruffano et al, 2018). Jones et al (2022) conducted a systematic review encompassing all studies related to AI technologies aimed at facilitating the early detection of skin cancer within primary and community care settings. The review included studies conducted from January 1, 2000 to August 9, 2021 (Jones et al, 2022). The authors categorized study populations into high-prevalence and low-prevalence populations and found that only 2 studies were conducted in low-prevalence populations. Hence, they opted to review the data from all 272 studies that employed AI and machine learning to assess skin lesions because these studies remain relevant to the implementation of these technologies in primary care. Although diagnostic accuracy was reasonable for melanoma (89.5%), SCC (85.3%), and BCC (87.6%), methodological heterogeneity and lack of data from low-prevalence clinical settings suggest caution in recommending widespread adoption of AI in primary care. The authors stressed that the populations under examination did not adequately reflect the diversity of the broader population. Furthermore, they highlighted challenges such as limited access to nonpublic datasets, insufficient transparency regarding training methodology, lack of prospective studies, the potential for bias and overfitting, and concerns regarding AI performance with out-of-distribution images (Dick et al, 2019; Jones et al, 2022).

The first randomized controlled trial in this field aimed to investigate whether a multiclass AI algorithm could enhance the accuracy of nondermatologists examining patients with suspicious skin lesions (Han et al, 2022). These skin lesions were identified either by the patient or a clinician without the use of dermoscopy. A total of 524 biopsy-proven cases and 52 clinically diagnosed cases were randomly selected for evaluation with or without AI support involving first-year dermatology residents and nondermatology trainees. The AI-aided group achieved a top 1 accuracy of 53.9%, contrasting with the unaided group's accuracy of 43.8%. The improvement was statistically significant among nondermatology trainees with limited dermatology experience, whereas it was not significant for dermatology residents.

Escalé-Besa et al (2023) conducted a prospective study to compare the diagnostic accuracy of AI with that of general practitioners (GPs) in a real-life setting. Eleven GPs were tasked with making a diagnosis and then subjecting the image to an AI model. Three questions probed the GPs on whether the model's results aided in diagnosis and management. The scope of this study extended beyond skin tumors to include various inflammatory conditions, which the AI was not adequately trained for, such as granuloma annulare, scabies, and hidradenitis. Despite adjustments for inadequate training, the diagnostic accuracy of the AI was lower than that of GPs and dermatologists (Escalé-Besa et al, 2023).

Regarding satisfaction and the acceptance of AI, Escalé-Besa et al (2023) found that 92% of GPs responded affirmatively to the question of whether AI aided in their differential diagnosis approach. In 60% of cases, the AI tool was instrumental in arriving at the diagnosis, and in 34% of cases, teledermatology consultation could have been avoided. Two survey-based studies yielded similar results, with GPs expressing strong support for AI and perceiving significant advantages, contrary to dermatologists' concerns about replacement (Samaran et al, 2021; Sangers et al, 2023).

AI AS A TOOL FOR DERMATOLOGISTS

Within dermatology, AI finds application across diverse tools dedicated to the detection of skin cancer, including TBP and dermoscopy. Market-approved AI-powered software for dermatologists, such as the MoleAnalyzer Pro (FotoFinder ATMB) and DEXI (Vectra WB360), are typically integrated with specialized hardware. These AI-powered apps have undergone testing in retrospective and prospective trials (Cerminara et al, 2023; Haenssle et al, 2020; Jahn et al, 2022; Winkler et al, 2023).

When applied to TBP for patients with multiple nevi, AI assumes a supportive role for dermatologists. In such cases, AI assists in sorting and visualizing nevi, empowering dermatologists to effectively compare images and identify changes and outliers. AI further facilitates the swift recognition of new or evolving lesions (Figure 1) by juxtaposing total-body images from current and previous visits (Salerni et al, 2012).

The utility of AI in determining total nevi count, a recognized risk factor for melanoma development, remains a subject of debate. An Australian study showcased the efficacy of AI in accurately quantifying total nevi count, revealing a substantial 97% concordance between AI-based counts and those conducted by senior dermatologists (Betz-Stablein et al, 2022).

However, a recent Swiss study reported significant differences between nevi counts determined by senior dermatologists and counts determined by AI. Although the AI-based lesion detection from 3-dimensional (3D) images exhibited performance superior to that from 2-dimensional images, both systems exhibited notable limitations for practical use in clinical routine owing to their tendency to overcount lesions (Cerminara et al, 2023).

Dermoscopy has been shown to enhance both the sensitivity and specificity of dermatologists in the detection of skin cancer detection (Kittler et al, 2002; Vestergaard et al, 2008). A recent systematic review by Haggenmüller et al (2021) assessed reader studies featuring direct comparisons between AI classifiers and human experts, underscoring the potential utility of AI-based classifiers in clinical practice. Of the 11 studies that focused on classification of dermoscopic images, the majority employed a binary classification system, distinguishing either between melanoma and nevus or between malignant and benign lesions. Importantly, all 11 studies were conducted within an artificial framework that deviated significantly from real-world scenarios.

In 7 of 11 studies, AI demonstrated diagnostic accuracy superior to that of dermatologists. AI has also shown superior performance in multiclass classification tasks, which encompass a broader range of clinically relevant diagnoses (Maron et al, 2019; Tschandl et al, 2019a). However, in the case of amelanotic lesions, AI exhibited similar or even lower performance than human raters (Tschandl et al, 2019b). Despite these striking results, the research community's focus is shifting. The man-against-machine scenario (Haenssle et al, 2018) is gradually being replaced by the more relevant human−computer collaboration, as pioneered by Tschandl et al (2020).

In a prospective diagnostic study by Winkler et al (2023), dermatologists were provided with AI malignancy scores, which they used to revise their initial diagnoses. Sensitivity and specificity significantly improved when dermatologists integrated AI-support results into their decision making. Dermatologists with less dermoscopy experience had the greatest improvement, whereas more experienced dermatologists showed little or no improvement (Tschandl et al, 2020; Winkler et al, 2023). Tschandl et al (2020) further showed that decision support with AI-based multiclass probabilities improved the accuracy of human raters from 63.6% to 77.0%. However, no improvement was observed for decision support with AI-based prediction of malignancy. This suggests that multiclass probabilities are the best form of AI output for the given task. Tschandl et al (2020) also demonstrated that even experts can be vulnerable to performing below their expected ability if they are wrongly influenced by a faulty AI.

A recent prospective study by Menzies et al (2023) compared novices, expert dermatologists, and an AI-powered smartphone app in a real-world clinical setting. They found that AI was superior to novices and on par with expert dermatologists for diagnosing pigmented skin lesions on dermoscopy images. However, in a setting involving patients with multiple nevi, AI was inferior to expert

Figure 1. AI in TBP. Comparison of 2 TBP sequential images in a patient aged 36 years with a personal history of stage Ia melanoma. AI quickly detected the new lesion (blue arrow) appearing on the back after a 6-month follow-up. The subject in Figure 1 consented to the publication of the image. AI, artificial intelligence; TBP, total body photography.

dermatologists in terms of specificity and management decisions. The study also showed that an AI algorithm that outperformed expert dermatologists in an artificial setting was inferior to experts in a real-world setting, highlighting the need for more prospective studies in the real world. Similarly, Combalia et al (2022) found that AI performed poorly compared with experts for out-of-sample dermoscopic images and for categories of lesions that were not included in the training set.

In the near future, collaborations between dermatologists and AI will be crucial. The realization of fully autonomous diagnostic machines, similar to self-driving cars, still seems distant and hypothetical for several reasons. First, the paucity of prospective trials in real-world settings precludes the implementation of stand-alone devices without an expert in the loop. Second, population screening with AI by TBP, although theoretically possible, may pose significant risks. This is because AI has lower specificity when applied to unselected individuals. The value of AI for the screening and surveillance of high-risk individuals after careful selection by expert clinicians remains to be determined (Russo et al, 2022). Finally, dermatologists tend to consider contextual information beyond morphology owing to their general knowledge. They also include human preferences in their decision-making process, such as the optimal trade off between sensitivity and specificity. These preferences may differ between different types of patients. Barata et al (2023) recently showed that including these preferences could improve AI-powered decision support. However, these preferences are not usually considered during AI training.

AI SUPPORT FOR OTHER NONINVASIVE IMAGING TECHNIQUES

AI has been applied to other noninvasive diagnostic techniques such as reflectance confocal microscopy (RCM) and optical coherence tomography (OCT). RCM allows the examination of the skin at cellular resolution from the surface down to the upper dermis, making it a suitable bridge between dermoscopy and histopathology. In several skin cancer referral centers where RCM is routinely used, the number of unnecessary biopsies has been reduced by up to 50% (Pellacani et al, 2014). However, RCM requires training, and training programs are scarce. In addition, the interpretation of RCM images is subjective, and there are no clearly defined diagnostic criteria for many skin conditions (Malciu et al, 2022). Machine learning algorithms offer a quantifiable and objective approach to RCM interpretation. They can also remove artifacts and reduce evaluation times. AI has been used to identify the dermal−epidermal junction (DEJ), evaluate the quality of RCM mosaics, differentiate between different types of cutaneous tumors, and differentiate between lentigo maligna and atypical melanocytic proliferations (Aractingi and Pellacani, 2019).

OCT generates high-resolution en face and cross-sectional images of the skin to a maximum depth of 2 mm, penetrating deeper than RCM. However, owing to its poorer resolution, the diagnosis of melanoma versus nevi is very challenging. Therefore, OCT is mainly used for the diagnosis of keratinocyte skin cancer. In an attempt to combine the advantages of RCM (cellular resolution) and OCT (penetration depth), line-field confocal OCT (LC-OCT) has been developed. LC-OCT provides 3 in vivo imaging modalities: vertical, horizontal, and a 3D reconstruction (Soglia et al, 2022). As in RCM, the identification of the DEJ can be challenging and can be improved by AI support (Chou et al, 2021). LC-OCT was also tested to differentiate between actinic keratosis and in situ SCC, but it turned out that the identification of cellular atypia by humans is very subjective, with poor inter-rater agreement. AI, on the other hand, could objectively define and quantify cellular atypia (Fischman et al, 2022).

A recent systematic review encompassed all currently applied AI methods for automated detection and classification of BCC using dermoscopy, RCM, and OCT. The reviewed publications demonstrate the potential benefit of AI in the detection of BCC, but some limitations are highlighted, such as lack of reproducibility of hand-crafted features, lack of transparency, and lack of publicly available large datasets (Widaatalla et al, 2023).

AI LIMITATIONS

In addition to offering significant opportunities, AI applications also confront a range of limitations and challenges, as outlined in Table 1. One notable challenge revolves around interpretability of deep learning−based AI algorithms. These algorithms, owing to their opaque nature, are often referred to as black boxes, rendering the interpretation of their predictions challenging (Castelvecchi, 2016; Salahuddin et al, 2021). To tackle this issue, the concept of explainable AI has been proposed as a potential solution. It is however still unclear whether explainable AI will have a meaningful impact on the decisions of clinicians (Hauser et al, 2022).

Another noteworthy limitation pertains to the generalizability of AI. AI models might struggle when confronted with images that deviate from those present in their training datasets. The performance of AI can be affected by minor alterations in input images, such as differences in magnification, differences in brightness, or the presence of artifacts such as markings or scale bars (Geirhos et al, 2020). The International Skin Imaging Collaboration AI Working Group identified specific deficiencies and safety concerns within AI diagnostic systems. In the test set for the International Skin Imaging Collaboration 2019 Grand Challenge, the organizers included low-quality images, images from diverse sources, images of lesions with diagnoses not represented in the training dataset, and images containing artifacts. These additional complexities contributed to a significant decline in diagnostic accuracy: the best-performing AI algorithms exhibited a reduction of >20% in the International Skin Imaging Collaboration than the best algorithms in the 2018 challenge (Combalia et al, 2022). Moreover, most existing algorithms focus solely on image data, neglecting valuable clinical metadata such as patient history, age, and risk factors. This omission hampers the algorithm's ability to generalize across diverse patient populations and clinical settings.

To address these challenges and limitations, the International Skin Imaging Collaboration AI Working Group has released a comprehensive checklist detailing best practices for the development and evaluation of image-based AI applications in dermatology. This checklist not only tackled data-related issues such as image acquisition and metadata

Table 1. Challenges and Opportunities of Implementing AI in Skin Cancer Diagnosis

Feature	Challenges	Opportunities
Selection of patients	Potential bias toward patients who are more tech savvy or have easier access to healthcare facilities equipped with AI technology	Easier and quicker screening options could democratize access to preliminary skin cancer diagnosis, benefiting patients in remote or underserved locations
Image acquisition	Lack of standardization in imaging protocols; variations in lighting, angle, and quality can affect diagnostic accuracy	Advances in imaging technology coupled with AI could lead to more standardized, high-quality imaging protocols
Diversity	Underrepresentation of skin types and conditions in training data can lead to biased algorithms that may not be universally applicable	By actively seeking diverse training datasets, AI models can be designed to be inclusive and accurate for a wide range of skin types and conditions
Privacy	Handling and storage of sensitive patient data, including images and medical history, may pose privacy concerns	AI can potentially automate the anonymization and secure storage of medical data, improving overall data privacy
Transparency and interpretability	The opacity of machine-learning algorithms makes it difficult for clinicians to interpret the reasoning behind a particular diagnosis, which could affect trust and accountability	Progress in explainable AI could make these systems more transparent, building trust among clinicians and patients
Implementation in routine	Integration into existing clinical workflows, compatibility with existing medical record systems, and staff training are significant hurdles	AI could automate repetitive tasks, allowing healthcare professionals to focus on more complex and patient-centered activities
Fairness	Ensuring that the AI tool does not discriminate on the basis of ethnicity, socioeconomic status, or other factors is difficult but crucial for equitable health care	Ethical AI practices and fairness-aware algorithms can help eliminate biases in health care
Accuracy	Although AI shows promise in controlled settings, its efficacy in varied, real-world conditions still requires extensive validation	In controlled settings, AI has shown remarkable accuracy, sometimes even outperforming human experts, thereby promising a future of improved diagnostic procedures
Regulations	The evolving landscape of healthcare regulations, particularly regarding AI and data usage, creates uncertainties that can slow down adoption and implementation	A well-defined regulatory framework can provide a trustworthy basis for implementing AI in health care, ensuring patient safety and data integrity

Abbreviation: AI, artificial intelligence.

but also ventured into technical aspects such as labeling and reference standards (Daneshjou et al, 2022a). Yet, another hurdle in the widespread adoption of AI for skin cancer diagnosis is the inherent bias of training data. Most AI systems have been trained on images of individuals with fair skin tones, thereby marginalizing those with darker skin tones. This lack of diversity in training can lead to subpar results when applying AI to individuals with darker skin tones (Daneshjou et al, 2022b). In addition, AI may falter when dealing with rare conditions, such as amelanotic melanoma or other rare diagnoses that were underrepresented in the training dataset (Tschandl et al, 2019b).

CONCLUSIONS

Overall, this overview of the evidence on the use of AI in skin cancer detection highlights a notable scarcity of well-designed prospective trials in real-life scenarios, thereby limiting definitive conclusions about its clinical utility. This limitation was particularly pronounced before 2020, the year in which the SPIRIT (Standard Protocol Items: Recommendations for Interventional Trials)-AI and CONSORT (Consolidated Standards of Reporting Trials)-AI guideline extensions were published. These extensions were specifically formulated to address criticisms related to evaluating interventions involving AI components (Liu et al, 2020; Cruz Rivera et al, 2020).

The current adoption of AI in skin cancer diagnosis is context dependent. For instance, in the domain of identifying changing or newly emerging lesions on whole-body photographs, AI has already seen widespread utilization. Conversely, in areas such as aiding nonexperts in diagnosing and treating skin lesions, AI's potential is largely untapped but promising. However, incorporating AI into consumer-focused smartphone apps presents a complex set of partly unknown risks and challenges. There remains a glaring absence of robust evidence supporting the benefits of these consumer-focused apps in practice. Similarly, the role of AI in primary care is a subject of active debate. Although GPs are increasingly acknowledging the technology's potential benefits, there is a pressing need to define its specific applications more clearly. Furthermore, the logistical challenge of seamlessly integrating AI into existing clinical workflows remains inadequately addressed. In summary, although the future of AI in skin cancer detection is replete with promise, we find ourselves merely at the inception of what is frequently predicted to be a paradigm-shifting transformation in healthcare.

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AUTHOR CONTRIBUTIONS

Conceptualization: GA, HK, AB, SP, JM; Investigation: GB, HK; Supervision: GA, HK; Writing - Original Draft Preparation: GB; Writing - Review and Editing: HK, AB, SP, JM, GA

CONFLICT OF INTEREST

AB has served on the advisory board and received honoraria for lectures and/or research grants for Amgen, Abbvie, Boehringer Ingelheim, Bristol Myers Squibb, Janssen, Lilly, Novartis, and UCB. JM is a cofounder of Athena Tech. SP declares serving as a speaker for Almirall, BMS, Cantabria, ISDIN, La Roche Posay, Leo Pharma, MSD, Novartis, Pfizer, Roche, Regeneron, Sanofi, and Sunpharma; serving on the advisory board for Almirall, BMS, ISDIN, La Roche Posay, Leo Pharma, Novartis, Pfizer, Regeneron, Roche, Sanofi, and Sun Pharma; research and trials for Abbvie, Almirall, Amgen, BMS, Biofrontera, Canfield, Cantabria, Fotofinder, GSK, ISDIN, La Roche Posay, Leo

Pharma, MSD, MEDA, Novartis, Polychem, Roche, and Sun Phaarma; and spouse interests in Almirall, Amgen, BMS, Biofrontera, Canfield, Cantabria, Fotofinder, GSK, ISDIN, La Roche Posay, Leo, Mavig, Nevisence, Novartis, Polychem, Roche, and Sun Pharma. Other declarations are educational activities for Abbie and Lilly, ISD, and Athena Tech & Dermavision Solutions. HK received speaker honorariums from Fotofinder; equipment from Fotofinder, Heine, 3-Gen, DermaMedical, and Casio; and license fees from MetaOptima, Casio, and Heine.

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